

Information Geometric Regularization for Euler Equations in Discontinuous Galerkin Methods

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MFEM Workshop 2026

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Compressible Euler Equations

- Governing hyperbolic conservation laws for **mass, momentum, and energy** in compressible flow.

$$\partial_t \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I} \\ (E + p) \mathbf{u} \end{pmatrix} = 0$$

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- Key challenge: smooth flow can rapidly compress and form shocks in finite time.

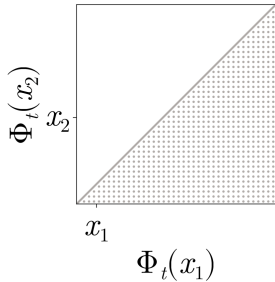
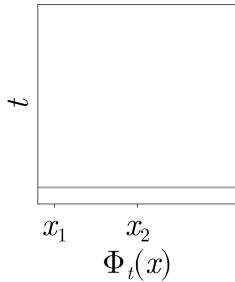
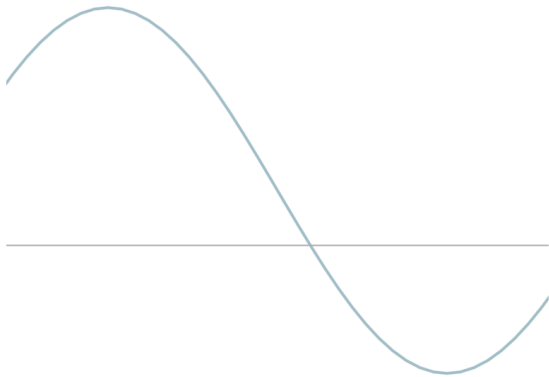
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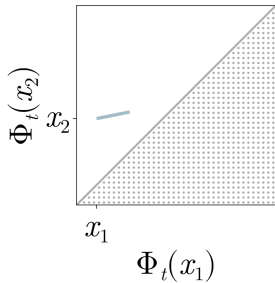
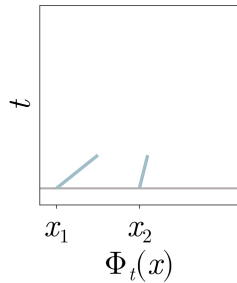
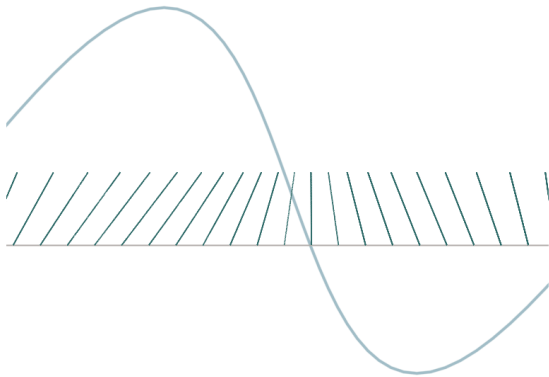
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- Key challenge: smooth flow can rapidly compress and form shocks in finite time.
- After shocks form, classical smooth solutions no longer exist.

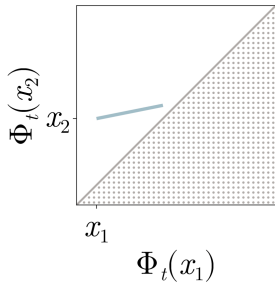
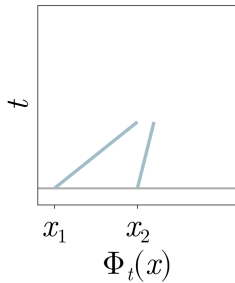
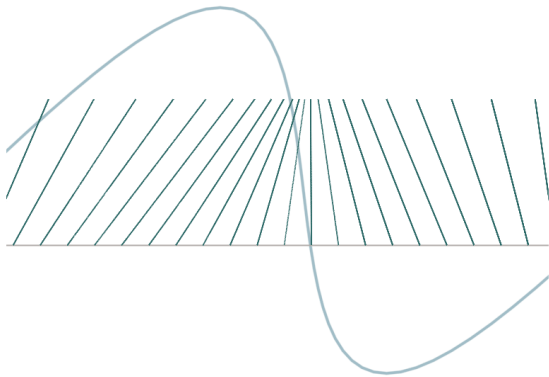
Trajectory Evolution



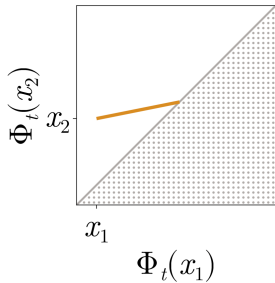
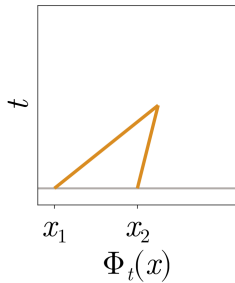
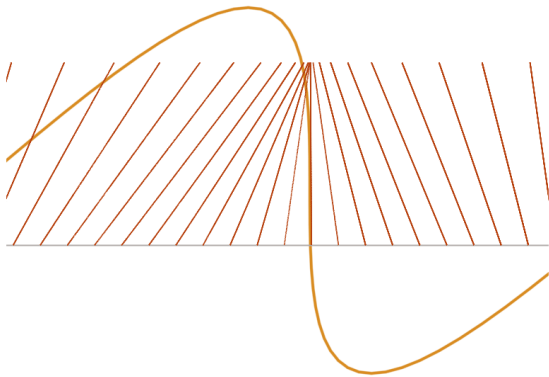
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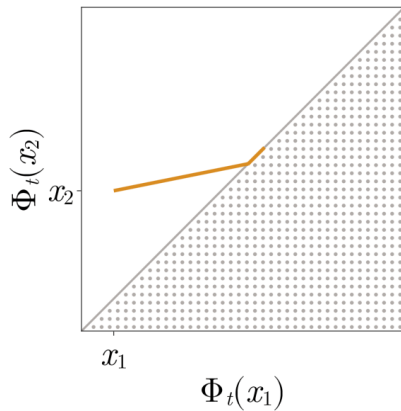
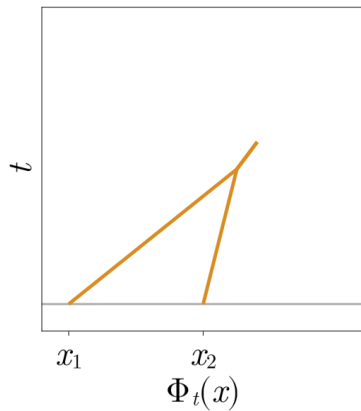
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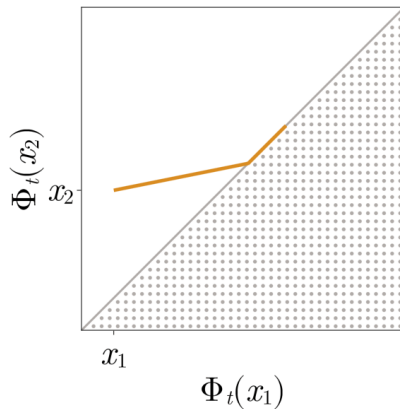
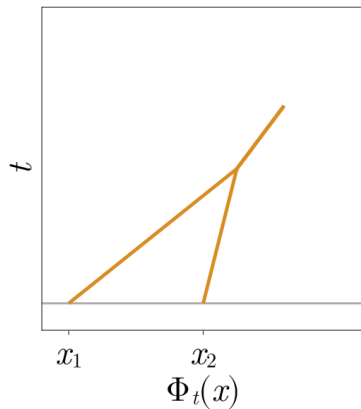
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Limits to Standard Shock Stabilization

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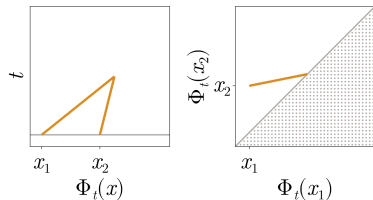
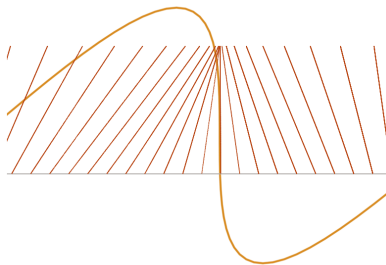
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- WENO methods improve shock capture but increase reconstruction complexity and are challenging to implement on non-uniform grids.
- Artificial viscosity improves stability but can require significant diffusion in high Mach flows.

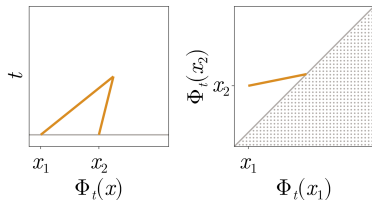
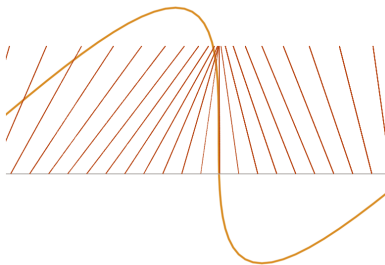
A Geometric Constraint on Compression

- Shock formation originates from compressive trajectories intersecting.



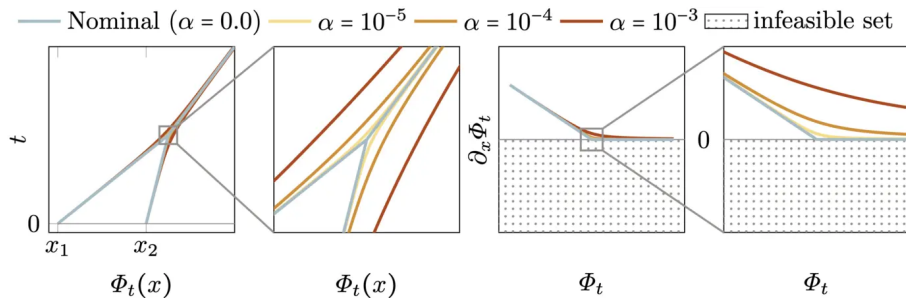
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- Existing stabilization methods suppress the solution at the discretization level after shocks form.
- Instead, introduce a geometric barrier term that prevents trajectories from intersecting.



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- The entropic pressure Σ is computed from an elliptic equation

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- Σ is recovered from the instantaneous flow state.

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Euler Discontinuous Galerkin (DG) Method

- Approximate the conserved variables with discontinuous basis functions

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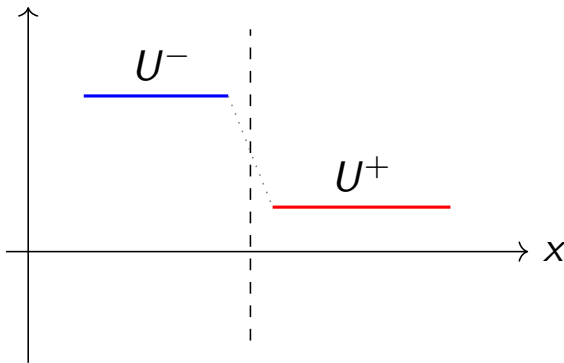
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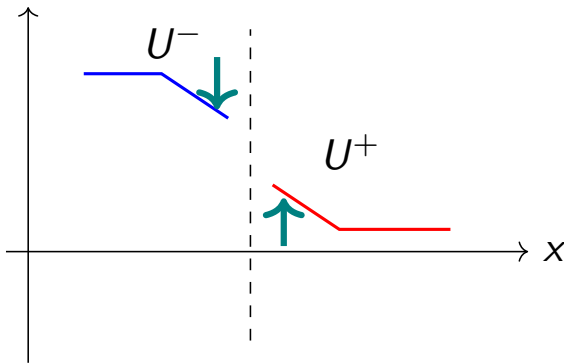


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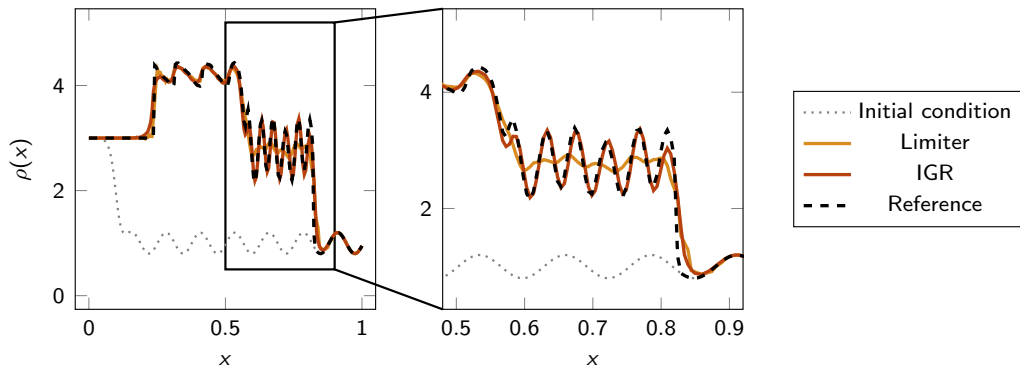
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- A is a sparse SPD matrix solved efficiently with a few iterative steps.

IGR Preserves Fine-Scale Structure

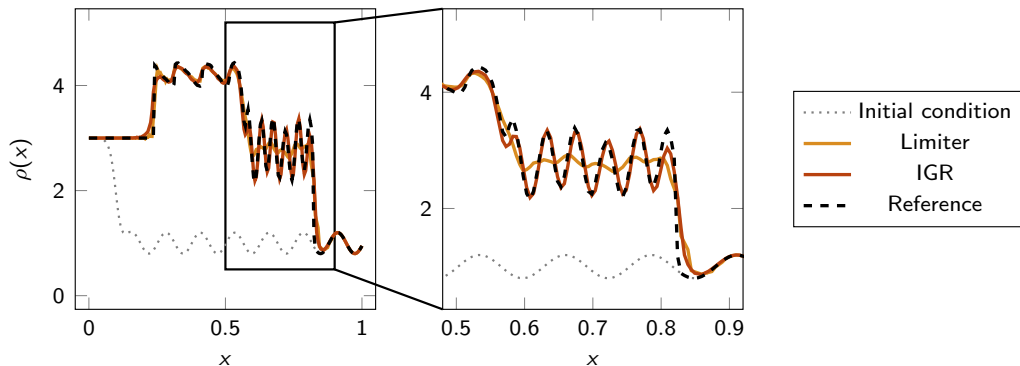
- Shock interacting with oscillatory structure



Smoothed Shu-Osher problem with $m = 64$, $p = 3$, and $t = 0.3$.

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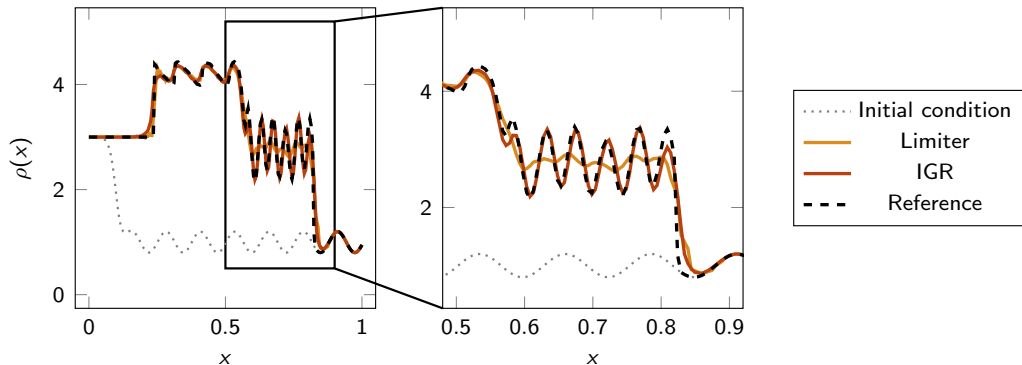
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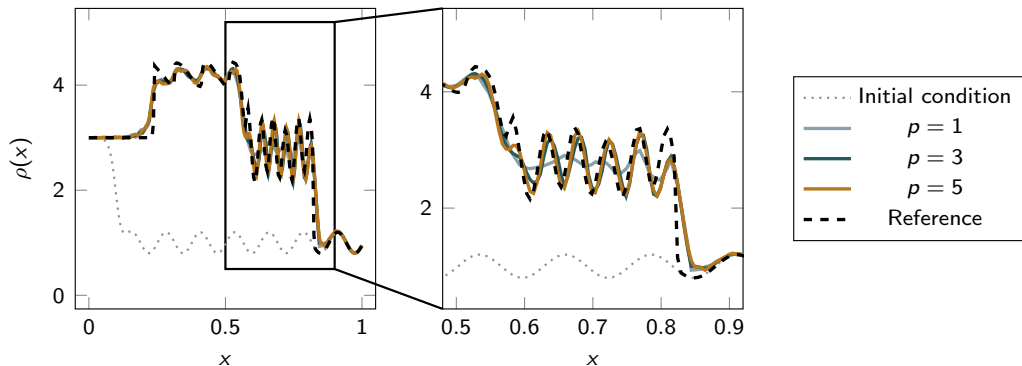
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- Limiter introduces damping of higher order modes and fine scale physical oscillations



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Effect of Polynomial Order

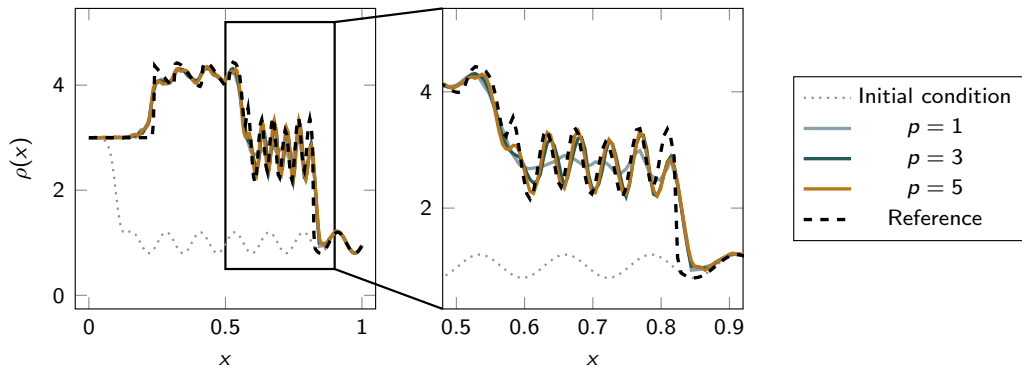
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Smoothed Shu-Osher problem at fixed $\text{DOF} = 128$ and $t = 0.3$.

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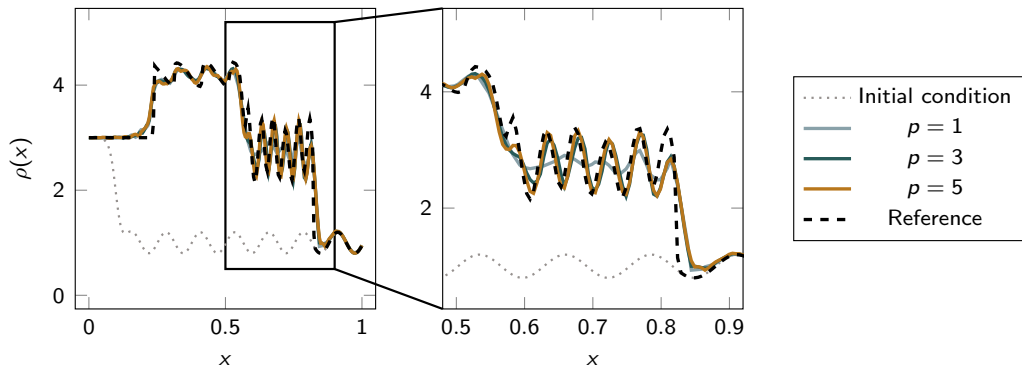
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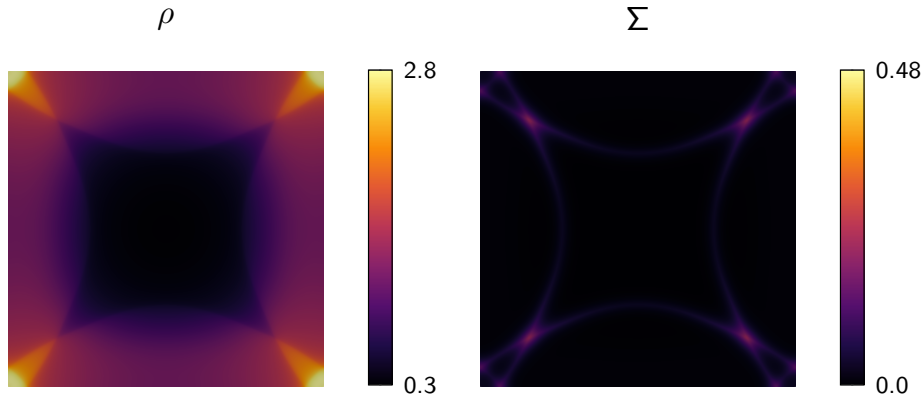
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- IGR retains fine-scale structure as p increases



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2D MFEM Implementation: Sedov Blast Wave

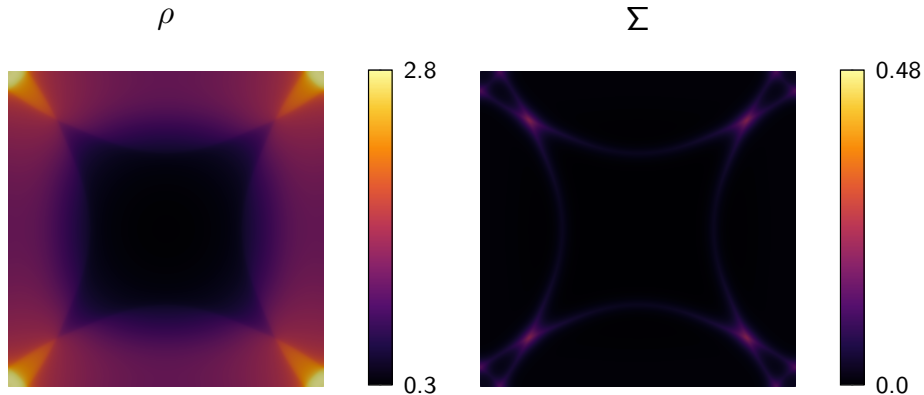
- Σ localizes near the expanding shock front



Two-dimensional Sedov blast wave using $m = 128 \times 128$ cells and $p = 3$.

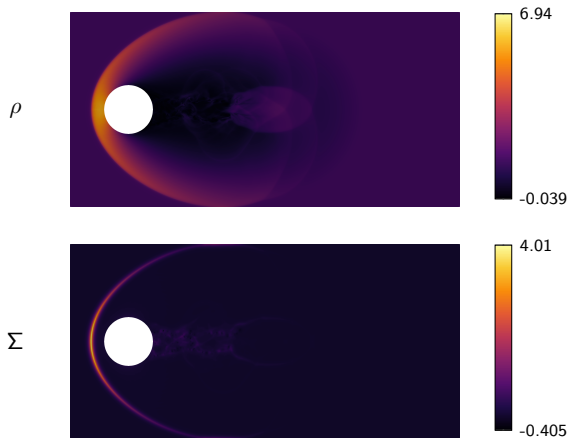
2D MFEM Implementation: Sedov Blast Wave

- Σ localizes near the expanding shock front
- Radial symmetry preserved without visible grid artifacts



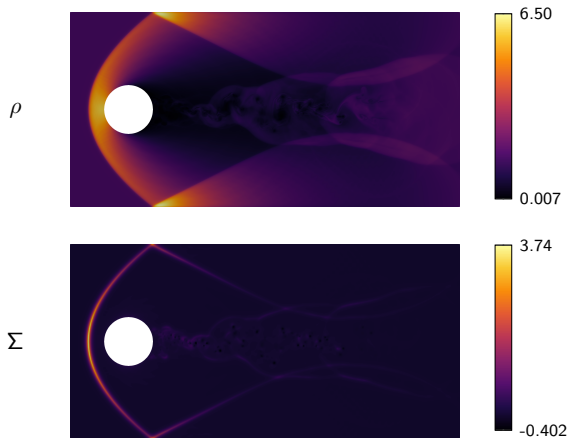
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2D MFEM Implementation: Supersonic Cylinder



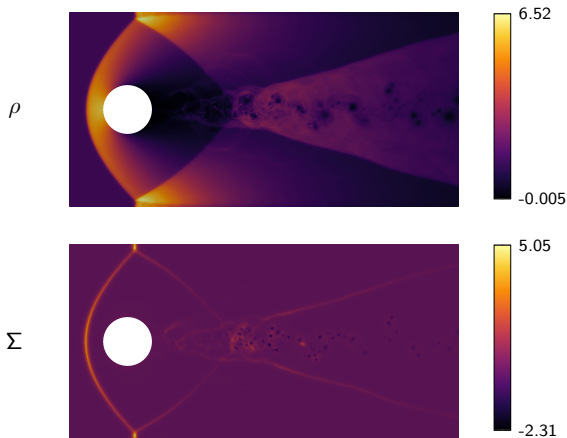
Two-dimensional Supersonic cylinder interaction using $m = 128 \times 128$ cells and $p = 2$.

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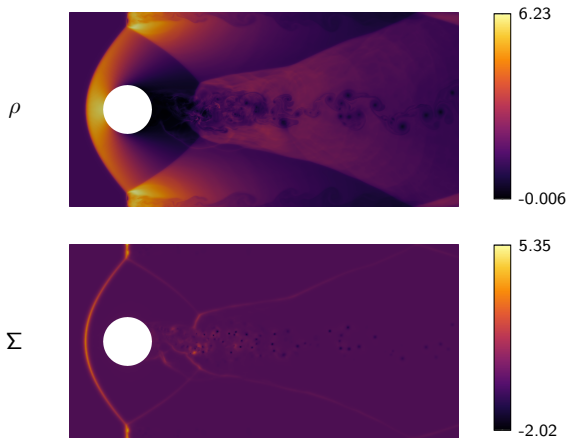
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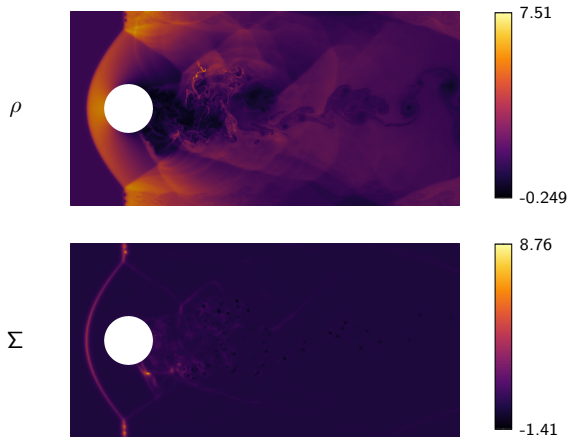
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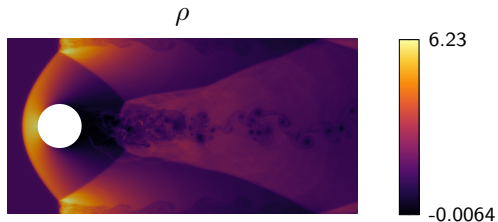
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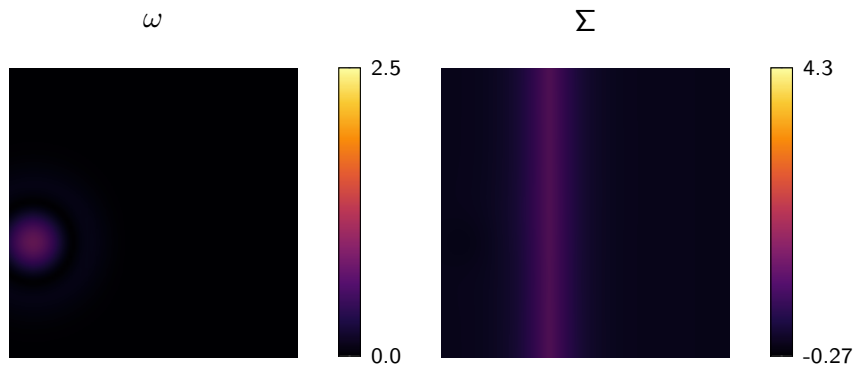
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- Preprint available at [arXiv:2608.02223](https://arxiv.org/abs/2608.02223) (2026).

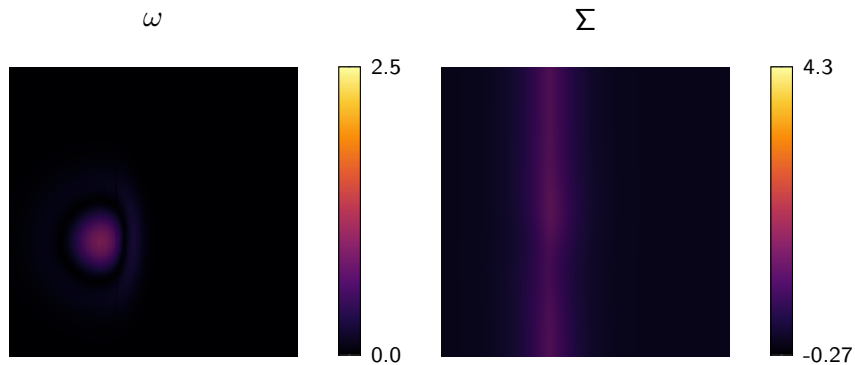
Backup Slides

2D MFEM Implementation: Shock–Vortex Interaction



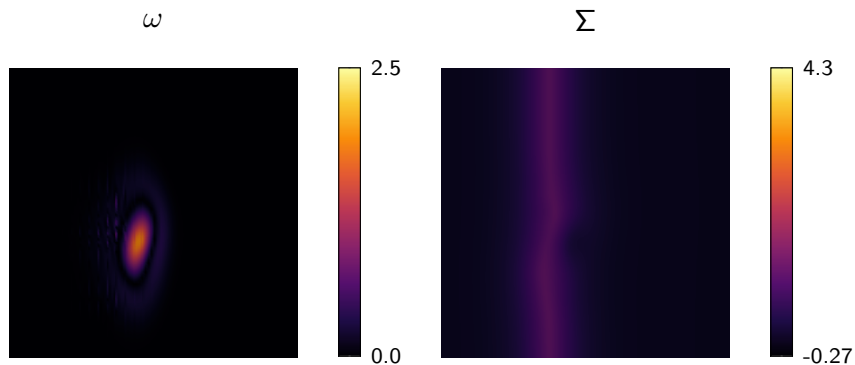
Zoomed shock–vortex interaction region using $m = 128 \times 128$ cells and $p = 3$.

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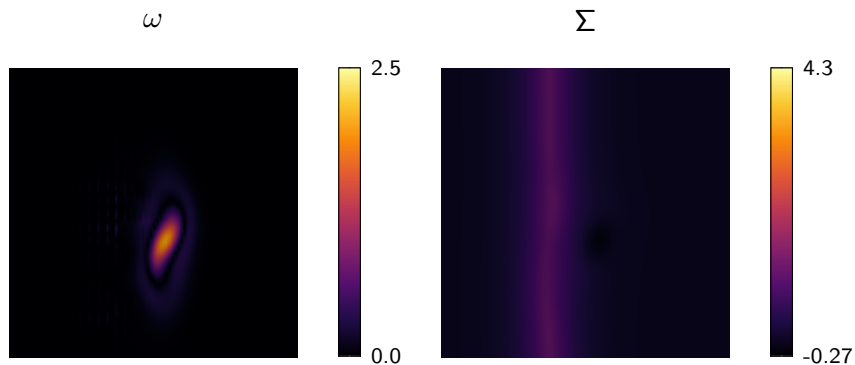
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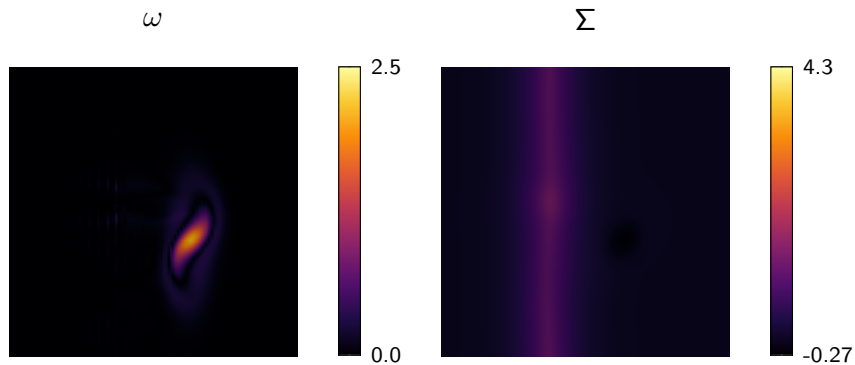
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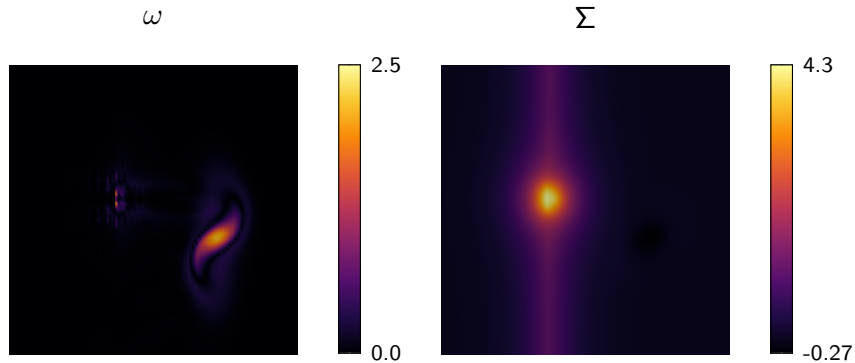
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